

# Association Rule Learning on the Groceries Dataset

Assignment: Association Rule Learning

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MSDA 9213 - Data Mining

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Code repository:

<https://github.com/ishejaflorent19-creator/Datamining-Assignement->

*Software: Python 3 (pandas, mlxtend, matplotlib)    /    Algorithm: Apriori*

September 2026

## Objective

This report applies association rule learning (the Apriori algorithm) to the Groceries transaction dataset to uncover which products customers tend to purchase together, and to interpret the resulting itemsets and rules.

## 1 Understand the Data

- a. **Number of transactions:** the dataset contains **9,835 transactions** (one grocery basket per row).
- b. **Number of distinct items:** there are **169 unique grocery items** across all transactions.
- c. **Ten most frequently purchased items**, with raw counts and support (fraction of transactions containing the item):

| Item             | Count | Support |
|------------------|-------|---------|
| whole milk       | 2513  | 0.2555  |
| other vegetables | 1903  | 0.1935  |
| rolls/buns       | 1809  | 0.1839  |
| soda             | 1715  | 0.1744  |
| yogurt           | 1372  | 0.1395  |
| bottled water    | 1087  | 0.1105  |
| root vegetables  | 1072  | 0.1090  |
| tropical fruit   | 1032  | 0.1049  |
| shopping bags    | 969   | 0.0985  |
| sausage          | 924   | 0.0940  |

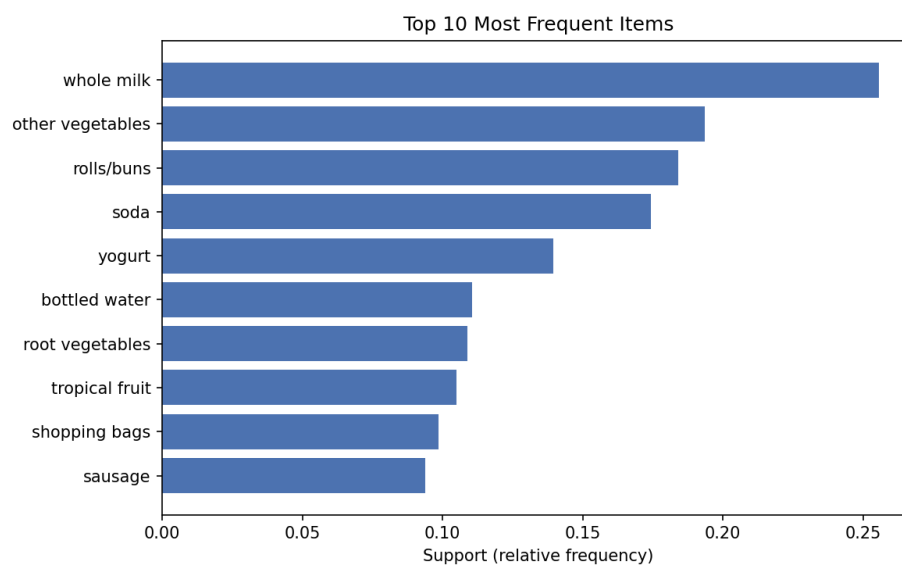


Figure 1: Top 10 most frequent items by support.

Whole milk and other vegetables are by far the most commonly purchased items, each appearing

in roughly a fifth to a quarter of all baskets, which is expected for staple grocery goods.

## 2 Frequent Itemsets

Using the Apriori algorithm with a minimum support threshold of 1% (0.01), **245 frequent itemsets** of size two or more were found. The 10 itemsets with the highest support are shown below.

| Itemset                           | Support |
|-----------------------------------|---------|
| other vegetables, whole milk      | 0.0748  |
| rolls/buns, whole milk            | 0.0566  |
| whole milk, yogurt                | 0.0560  |
| root vegetables, whole milk       | 0.0489  |
| other vegetables, root vegetables | 0.0474  |
| other vegetables, yogurt          | 0.0434  |
| other vegetables, rolls/buns      | 0.0426  |
| tropical fruit, whole milk        | 0.0423  |
| soda, whole milk                  | 0.0401  |
| rolls/buns, soda                  | 0.0383  |

**c. Highest-support itemset:** {other vegetables, whole milk}, support = 0.0748 (about 7.5% of all transactions).

**d. Interpretation:** a support of 0.0748 means that roughly 7 to 8 out of every 100 shopping trips include both other vegetables and whole milk. Because support is a population-wide frequency, high-support itemsets identify combinations that are common across the whole customer base, not just for a small niche of shoppers — these are good candidates for store layout and promotion decisions that need to affect many customers at once.

## 3 Association Rules

Rules were generated from the frequent itemsets using a minimum confidence threshold of 0.2, producing **234 rules** in total. The 10 rules with the highest lift are reported below (antecedent → consequent).

| Antecedent                       | Consequent                   | Support | Confidence | Lift  |
|----------------------------------|------------------------------|---------|------------|-------|
| citrus fruit, other vegetables   | root vegetables              | 0.0104  | 0.3592     | 3.295 |
| other vegetables, yogurt         | whipped/sour cream           | 0.0102  | 0.2342     | 3.267 |
| other vegetables, tropical fruit | root vegetables              | 0.0123  | 0.3428     | 3.145 |
| beef                             | root vegetables              | 0.0174  | 0.3314     | 3.040 |
| citrus fruit, root vegetables    | other vegetables             | 0.0104  | 0.5862     | 3.030 |
| root vegetables, tropical fruit  | other vegetables             | 0.0123  | 0.5845     | 3.021 |
| other vegetables, whole milk     | root vegetables              | 0.0232  | 0.3098     | 2.842 |
| root vegetables                  | other vegetables, whole milk | 0.0232  | 0.2127     | 2.842 |
| butter                           | other vegetables, whole milk | 0.0115  | 0.2073     | 2.771 |
| curd, whole milk                 | yogurt                       | 0.0101  | 0.3852     | 2.761 |

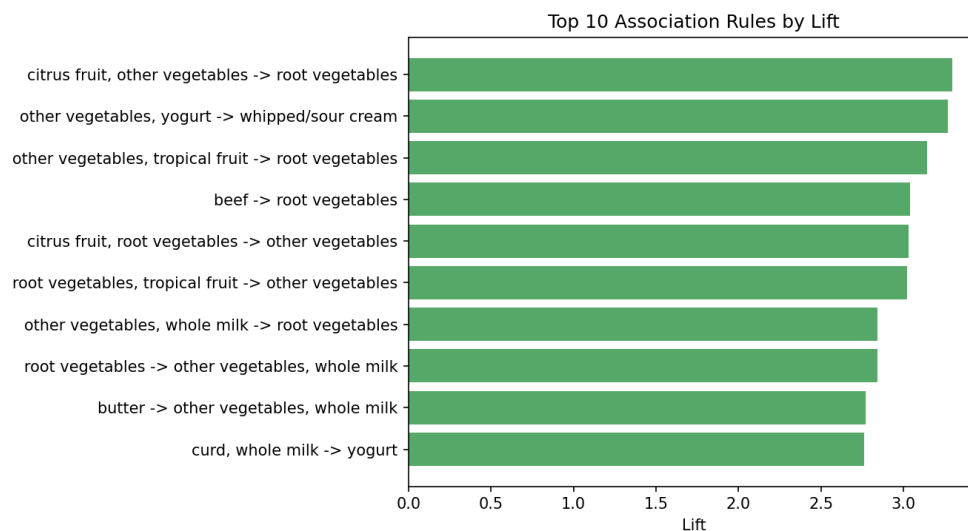


Figure 2: Top 10 association rules ranked by lift.

The rules cluster into two visible groups: a low-support/high-confidence/high-lift group (rare but strong fresh-produce combinations) and a high-support/lower-lift group (common staples such as whole milk and soda, which co-occur often simply because they are each bought so frequently).

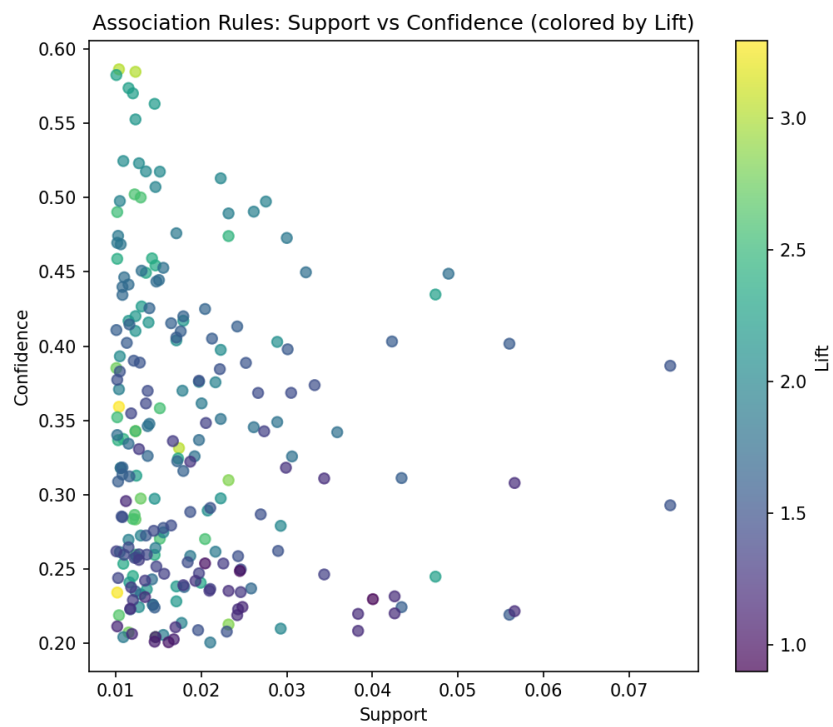


Figure 3: All 234 rules: support vs. confidence, colored by lift.

## 4 Understanding the Measures

Three rules from Section 3 are discussed in detail.

### Rule 1: {citrus fruit, root vegetables} $\rightarrow$ {other vegetables}

- **Support (0.0104):** about 1% of all transactions contain citrus fruit, root vegetables, and other vegetables together.
- **Confidence (0.5862):** among customers who bought citrus fruit and root vegetables, 58.6% also bought other vegetables.
- **Lift (3.03):** customers who buy citrus fruit and root vegetables are about 3 times more likely to also buy other vegetables than a random customer — a strong positive association.

### Rule 2: {root vegetables} $\rightarrow$ {other vegetables, whole milk}

- **Support (0.0232):** about 2.3% of all transactions contain root vegetables together with both other vegetables and whole milk.
- **Confidence (0.2127):** of the customers who bought root vegetables, 21.3% also bought both other vegetables and whole milk.
- **Lift (2.842):** buying root vegetables makes a customer about 2.8 times more likely to also buy this pair than an average customer.

### Rule 3: {whole milk} $\rightarrow$ {other vegetables} (highest-support rule)

- **Support (0.0748):** about 7.5% of all transactions contain both whole milk and other vegetables.
- **Confidence (0.2929):** of the customers who bought whole milk, 29.3% also bought other vegetables.
- **Lift (1.514):** a positive but comparatively weaker association than Rules 1 and 2, even though its support is far higher.

## 5 Find Interesting Rules

- Highest confidence:** {citrus fruit, root vegetables}  $\rightarrow$  {other vegetables}, confidence = 0.5862.
- Highest lift:** {citrus fruit, other vegetables}  $\rightarrow$  {root vegetables}, lift = 3.295.
- Highest support:** {whole milk}  $\rightarrow$  {other vegetables}, support = 0.0748.
- Are these the same rule?** No. This is expected: support favors rules built from very common items (whole milk, other vegetables) regardless of how strong the association actually is; confidence favors rules where the consequent is nearly guaranteed once the antecedent is present, even if that combination is rare overall; and lift favors rules where the antecedent and consequent occur together far more than chance would predict, independent of how frequent they are. A rule can therefore be popular (high support) without being a particularly strong or surprising association (low lift), and vice versa.

## 6 Interpret a Rule

**Selected rule:** {citrus fruit, root vegetables}  $\rightarrow$  {other vegetables}.

In simple language: customers who purchase citrus fruit and root vegetables also tend to purchase other vegetables. Specifically, about 58.6% of the baskets containing citrus fruit and root vegetables also contain other vegetables, and this combination is about 3 times more likely than chance (lift = 3.03).

This represents a **strong association**. The confidence is well above the 20% threshold used to generate the rules and above the baseline purchase rate of other vegetables (about 19.3% of all transactions), and the lift is comfortably greater than 1, indicating the co-occurrence is not simply because both items are individually popular. Together, high confidence and high lift indicate a genuine purchasing pattern — likely because these are all fresh-produce items bought together for the same meals — rather than a coincidence of two common items.

## 7 Business Application

- a. **Two products to place close to each other:** root vegetables and other vegetables (or citrus fruit and root vegetables) — both pairs show high confidence and high lift, so placing them in adjacent aisle sections would put related items in the customer’s path.
- b. **Possible product bundle:** a “fresh produce basket” bundle combining citrus fruit, root vegetables, and other vegetables, since all three co-occur frequently and with high lift.
- c. **Recommendation example:** when a customer adds root vegetables and citrus fruit to their (online or self-checkout) basket, the system could automatically suggest other vegetables — similar to an Amazon-style “customers who bought this also bought...” recommendation — since historical data shows a 58.6% chance the customer will want that item too.

## 8 Understanding Association

- a. **Does an association rule prove causation?** No. Association rules describe correlation in purchasing behavior, not causation. A high-confidence, high-lift rule such as {citrus fruit, root vegetables}  $\rightarrow$  {other vegetables} shows that these items are frequently bought together, but it does not show that buying citrus fruit and root vegetables *causes* someone to buy other vegetables. A third factor — for example, a general tendency toward cooking fresh, vegetable-based meals — could explain why all three appear together, which would be a confound rather than a causal chain.
- b. **Why might a rule have high confidence but low lift?** Confidence only measures how often the consequent appears given the antecedent, while lift compares that to how often the consequent appears overall. If the consequent (for example, whole milk) is purchased in a very large fraction of all transactions regardless of what else is bought, then almost any rule ending in “ $\rightarrow$  whole milk” will have high confidence simply because whole milk is everywhere — but the lift will be close to 1 (or even below it) because buying the antecedent did not really change the customer’s likelihood of buying whole milk versus the baseline.
- c. **What does a lift greater than 1 indicate?** A lift above 1 indicates a positive association: the antecedent and consequent occur together more often than would be expected if they

were purchased independently. The further above 1, the stronger the association (lift = 1 means independence, and lift below 1 would indicate a negative association, where buying the antecedent makes the consequent less likely).

## Conclusion

Applying the Apriori algorithm to the Groceries dataset (9,835 transactions, 169 items) surfaced clear and interpretable purchasing patterns. Fresh produce items — other vegetables, root vegetables, citrus fruit, and tropical fruit — form the strongest associations (lift around 3), while rules involving very high-frequency staples such as whole milk tend to have high support but comparatively modest lift. These findings translate directly into actionable retail decisions: co-locating strongly associated fresh-produce items, building cross-category bundles, and powering “customers who bought this also bought...” recommendations.

## A Analysis Code

The full Python code used for this analysis (data loading, frequent itemset mining, rule generation, and figure creation) is reproduced below. It uses `pandas` and `mlxtend`.

```
"""
Association Rule Learning on the Groceries Dataset
Apriori algorithm via mlxtend
"""

import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules
from mlxtend.preprocessing import TransactionEncoder
import matplotlib.pyplot as plt
import matplotlib
matplotlib.use("Agg")

# -----
# 1. Load data - the CSV is a "basket" file: one transaction per
# row, items comma-separated, no header, ragged row lengths.
# -----
DATA_PATH = "data/groceries.csv"

transactions = []
with open(DATA_PATH, "r", encoding="utf-8") as f:
    for line in f:
        items = [i.strip() for i in line.strip().split(",") if i.strip() != ""]
        if items:
            transactions.append(items)

n_transactions = len(transactions)
all_items = sorted(set(item for t in transactions for item in t))
n_items = len(all_items)

print(f"Number of transactions: {n_transactions}")
print(f"Number of distinct items: {n_items}")

# -----
# 2. Top 10 most frequent items
# -----
from collections import Counter
item_counts = Counter(item for t in transactions for item in t)
top10_items = pd.DataFrame(item_counts.most_common(10), columns=["item", "count"])
top10_items["support"] = top10_items["count"] / n_transactions
print("\nTop 10 items:\n", top10_items)

plt.figure(figsize=(8, 5))
plt.barh(top10_items["item"][:-1], top10_items["support"][:-1], color="#4C72B0")
plt.xlabel("Support (relative frequency)")
plt.title("Top 10 Most Frequent Items")
plt.tight_layout()
plt.savefig("figures/top10_items.png", dpi=150)
plt.close()

# -----
# 3. One-hot encode transactions
# -----
te = TransactionEncoder()
te_ary = te.fit(transactions).transform(transactions)
```

```

df = pd.DataFrame(te_ary, columns=te.columns_)

# -----
# 4. Frequent itemsets (Apriori), min_support chosen so we get a
# healthy number of itemsets of size >= 2
# -----
min_support = 0.01
frequent_itemsets = apriori(df, min_support=min_support, use_colnames=True)
frequent_itemsets["length"] = frequent_itemsets["itemsets"].apply(len)

itemsets_2plus = frequent_itemsets[frequent_itemsets["length"] >= 2].sort_values(
    "support", ascending=False
)
print(f"\nFrequent itemsets (size>=2) at min_support={min_support}: {len(itemsets_2plus)}")
print(itemsets_2plus.head(10))

top10_itemsets = itemsets_2plus.head(10).copy()
top10_itemsets["itemsets"] = top10_itemsets["itemsets"].apply(lambda s: ", ".join(sorted(s)))
top10_itemsets[["itemsets", "support"]].to_csv("results/top10_frequent_itemsets.csv", index=False)

# -----
# 5. Association rules
# -----
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.2)
rules = rules.sort_values("lift", ascending=False).reset_index(drop=True)

# keep readable columns
rules_readable = rules.copy()
rules_readable["antecedents"] = rules_readable["antecedents"].apply(lambda s: ", ".join(sorted(s)))
rules_readable["consequents"] = rules_readable["consequents"].apply(lambda s: ", ".join(sorted(s)))
cols = ["antecedents", "consequents", "support", "confidence", "lift"]
rules_readable = rules_readable[cols]

print(f"\nTotal rules generated (confidence >= 0.2): {len(rules)}")
top10_rules = rules_readable.head(10)
print(top10_rules)
top10_rules.to_csv("results/top10_rules_by_lift.csv", index=False)

# Rules sorted by confidence and support as well, for Q5
by_conf = rules_readable.sort_values("confidence", ascending=False).iloc[0]
by_lift = rules_readable.sort_values("lift", ascending=False).iloc[0]
by_supp = rules_readable.sort_values("support", ascending=False).iloc[0]

print("\nHighest confidence rule:\n", by_conf)
print("\nHighest lift rule:\n", by_lift)
print("\nHighest support rule:\n", by_supp)

with open("results/top_rules_summary.txt", "w") as f:
    f.write("HIGHEST CONFIDENCE RULE\n")
    f.write(str(by_conf) + "\n\n")
    f.write("HIGHEST LIFT RULE\n")
    f.write(str(by_lift) + "\n\n")
    f.write("HIGHEST SUPPORT RULE\n")
    f.write(str(by_supp) + "\n")

# -----
# 6. Scatter plot: support vs confidence, colored by lift
# -----
plt.figure(figsize=(7, 6))

```

```
sc = plt.scatter(rules["support"], rules["confidence"], c=rules["lift"], cmap="viridis", alpha=0.7)
plt.colorbar(sc, label="Lift")
plt.xlabel("Support")
plt.ylabel("Confidence")
plt.title("Association Rules: Support vs Confidence (colored by Lift)")
plt.tight_layout()
plt.savefig("figures/rules_scatter.png", dpi=150)
plt.close()

# Bar chart of top 10 rules by lift
plt.figure(figsize=(9, 5))
labels = [f"{a} -> {c}" for a, c in zip(top10_rules["antecedents"], top10_rules["consequents"])]
plt.barh(labels[:-1], top10_rules["lift"][:-1], color="#55A868")
plt.xlabel("Lift")
plt.title("Top 10 Association Rules by Lift")
plt.tight_layout()
plt.savefig("figures/top10_rules_lift.png", dpi=150)
plt.close()

frequent_itemsets.to_csv("results/all_frequent_itemsets.csv", index=False)
rules_readable.to_csv("results/all_rules.csv", index=False)

print("\nDone. Outputs written to current directory and figures/ subfolder.")
```